Chemistry / GCSE / IGCSE

Balancing chemical equations

A balanced equation has the same number of atoms of each element on both sides. Change coefficients in front of formulae.

Core ideas

1. Write the correct chemical formulae first.

2. Count each element on the reactant and product sides.

3. Adjust coefficients and recount. Use the smallest whole-number ratio. Do not change subscripts to balance an equation.

Worked example

For hydrogen reacting with oxygen, start $H_2 + O_2 \rightarrow H_2O$. Balance oxygen with $2H_2O$, then hydrogen with $2H_2$. The result is $2H_2 + O_2 \rightarrow 2H_2O$.

Try it yourself

1. Balance $Mg + O_2 \rightarrow MgO$.

2. Balance $N_2 + H_2 \rightarrow NH_3$.

3. Why must the subscript in H_2O stay unchanged?

Check your answers

1. $2Mg + O_2 \rightarrow 2MgO$

2. $N_2 + 3H_2 \rightarrow 2NH_3$

3. Changing a subscript changes the substance's formula.